

Vision Transformers

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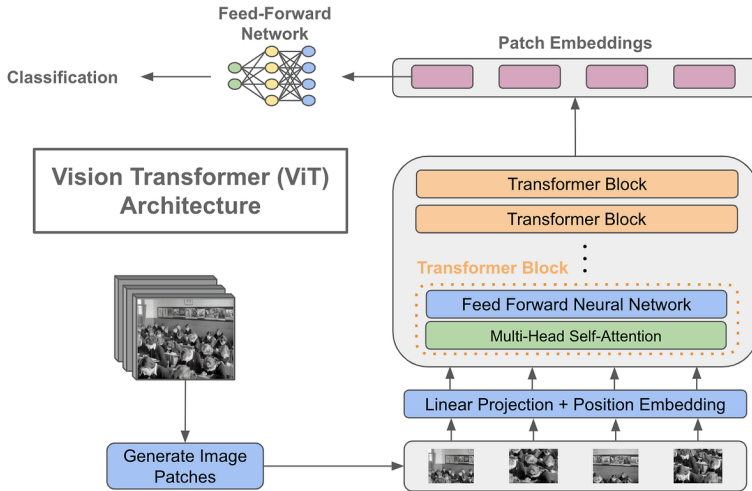
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Why this topic matters:

- ▶ Transformers dominate NLP; can the same idea work for images?
- ▶ Vision Transformers (ViTs) treat an image as a sequence of patches.
- ▶ With enough data, ViTs match or beat CNNs on many benchmarks.
- ▶ Hierarchical and hybrid designs make them practical for detection and segmentation.
- ▶ *“An image is worth 16×16 words.”* — Dosovitskiy et al., 2020.

By the end of this session, you will be able to:

- ▶ Explain how an image is converted into patch embeddings for a Transformer.
- ▶ Describe the ViT architecture (patches, class token, positional encodings, MLP head).
- ▶ Understand data-efficient training via distillation (DeiT).
- ▶ Compare CNNs and ViTs in terms of inductive bias and data efficiency.
- ▶ Describe hierarchical ViTs (Swin) and Transformer-based detection (DETR).
- ▶ Recognize that modern CNNs (ConvNeXt) remain competitive.

Vision Transformers: **The ViT Architecture**

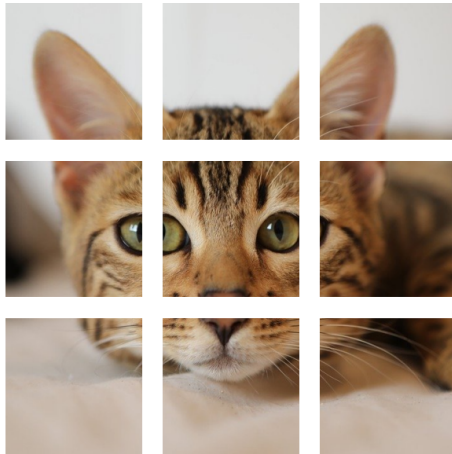
Key idea:

- ▶ Split image into fixed-size patches (e.g., 16×16)
- ▶ Flatten and project patches into embeddings
- ▶ Add positional encodings to embeddings
- ▶ Apply standard Transformer encoder

Architecture:

- ▶ Patch embedding
- ▶ Class token
- ▶ Transformer blocks
- ▶ MLP Head

Source: Dosovitskiy et al., "An Image is Worth 16×16 Words", 2020



Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 52

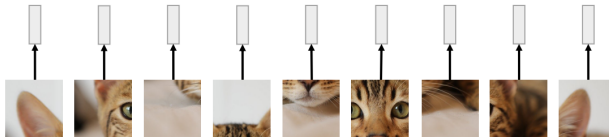
N input patches, each
of shape $3 \times 16 \times 16$



Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 53

Linear projection to
D-dimensional vector

N input patches, each
of shape $3 \times 16 \times 16$

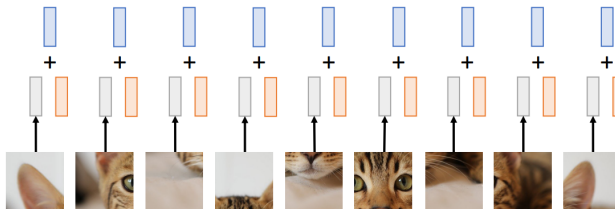


Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 54

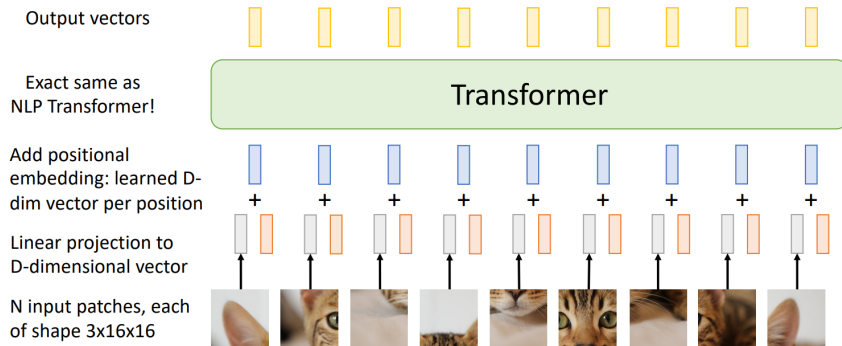
Add positional
embedding: learned D-
dim vector per position

Linear projection to
D-dimensional vector

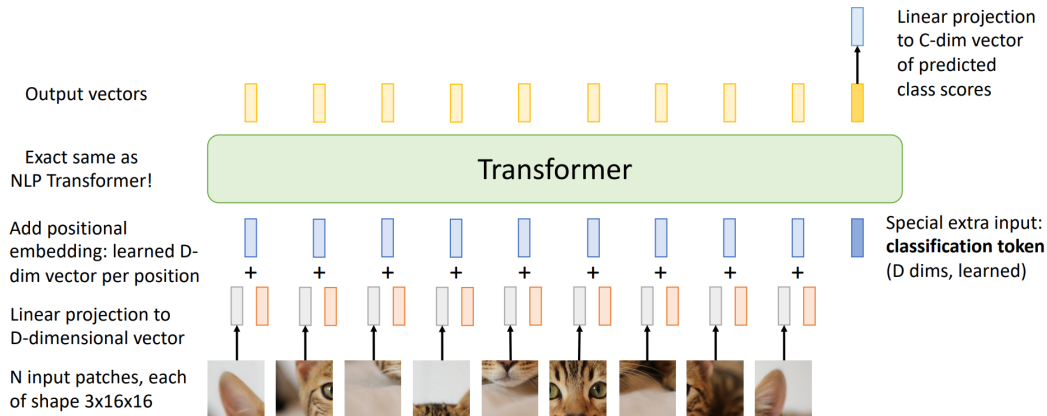
N input patches, each
of shape 3x16x16



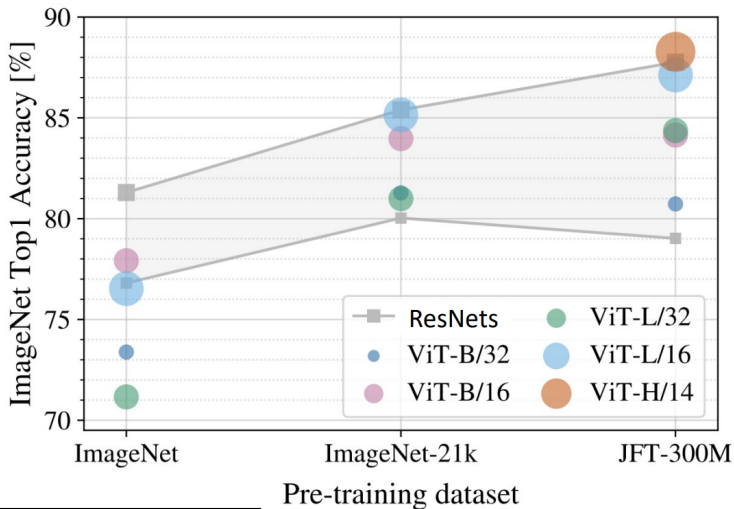
Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 55



Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 56

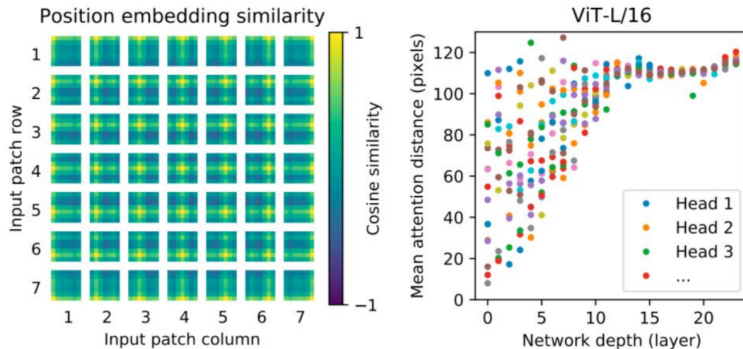


Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 58



⁰Dosovitskiy et al, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", ICLR 2021

- ▶ **Patch Embedding:** Achieved via a linear projection of flattened patches ($P = 16$ typical)
- ▶ **Differences from CNN stems:**
 - Large stride causes optimization instability
 - Mitigated via convolutional preprocessing (e.g., small conv stem)



Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", ICLR 2021, Fig. 7 (center and right panels)

- ▶ ViT learns the grid-like structure of the image patches via its position embeddings.
- ▶ The lower layers contain both global and local features, while the higher layers contain only global features.

Vision Transformers: **Improving ViT: Distillation**

Step 1: Train a teacher CNN on ImageNet

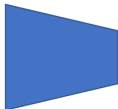


$P(\text{cat}) = 0.9$
 $P(\text{dog}) = 0.1$

Cross
Entropy
Loss

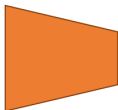
GT label:
Cat

Step 2: Train a student ViT to match ImageNet predictions from the teacher CNN (and match GT labels)



$P(\text{cat}) = 0.1$
 $P(\text{dog}) = 0.9$

KL Divergence Loss

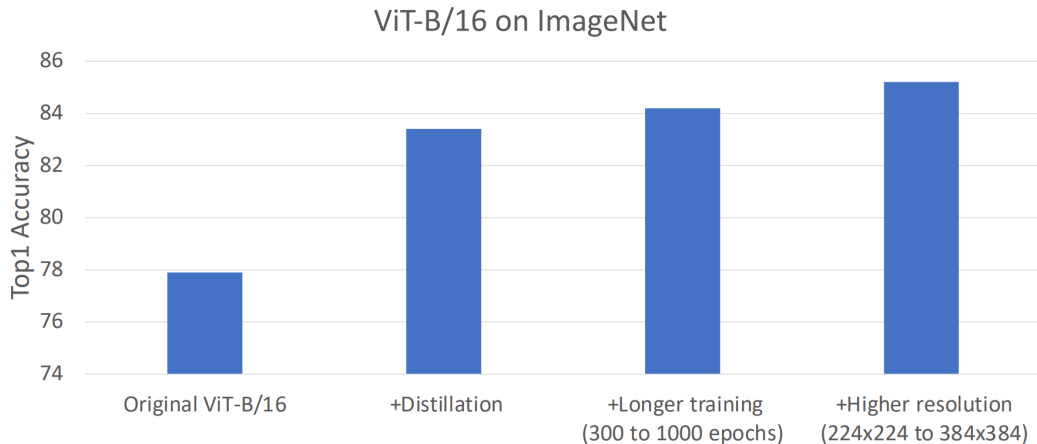


$P(\text{cat}) = 0.2$
 $P(\text{dog}) = 0.8$

Cross
Entropy
Loss

GT label:
Dog

Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 85



⁰Touvron et al., "Training data-efficient image transformers & distillation through attention", ICML 2021

Vision Transformers: **CNN** vs **ViT**

CNN



1. Maintain 2D structure logic



2. Shift equivariant



3. Consider only local correlations



4. Hierarchically growing field of view



5. Hierarchically progressing complexity



6. Reasonable amount of params

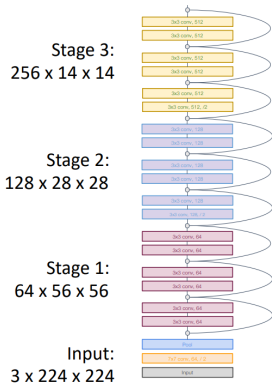


7. Global representation

ViT



Source: WAIC DL4CV, "From Sequences to ViT" (S. Bagon), slide 30

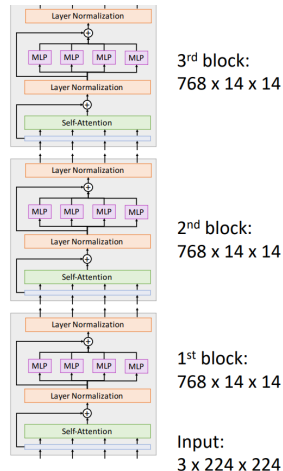


In most CNNs (including ResNets), **decrease** resolution and **increase** channels as you go deeper in the network (Hierarchical architecture)

Useful since objects in images can occur at various scales

In a ViT, all blocks have same resolution and number of channels (Isotropic architecture)

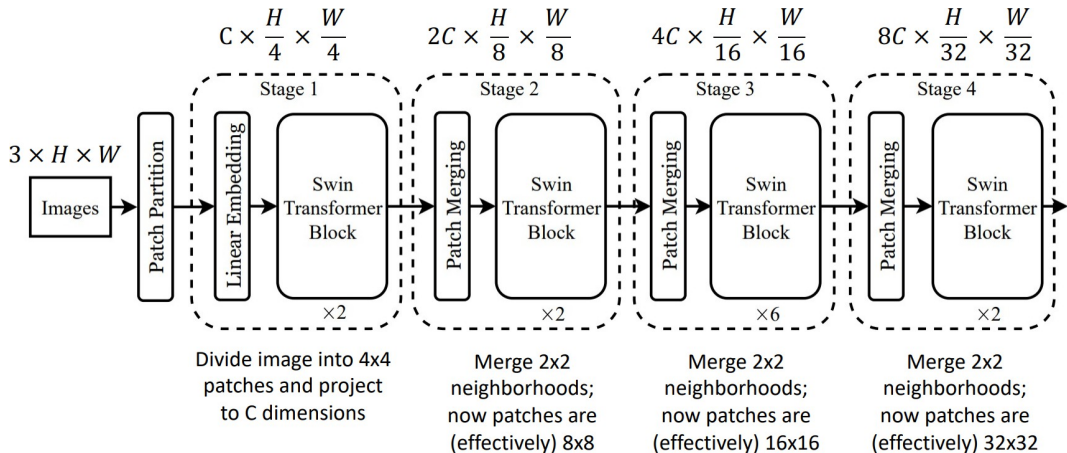
Can we build a **hierarchical** ViT model?



Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 91

Feature	CNNs	Vision Transformers
Locality	Strong inductive bias	Weak (learns from data)
Parameter sharing	Yes	Yes
Global context	Limited	Native support
Training data	Efficient on small data	Needs large data or pretraining
Flexibility	Less modular	More modular

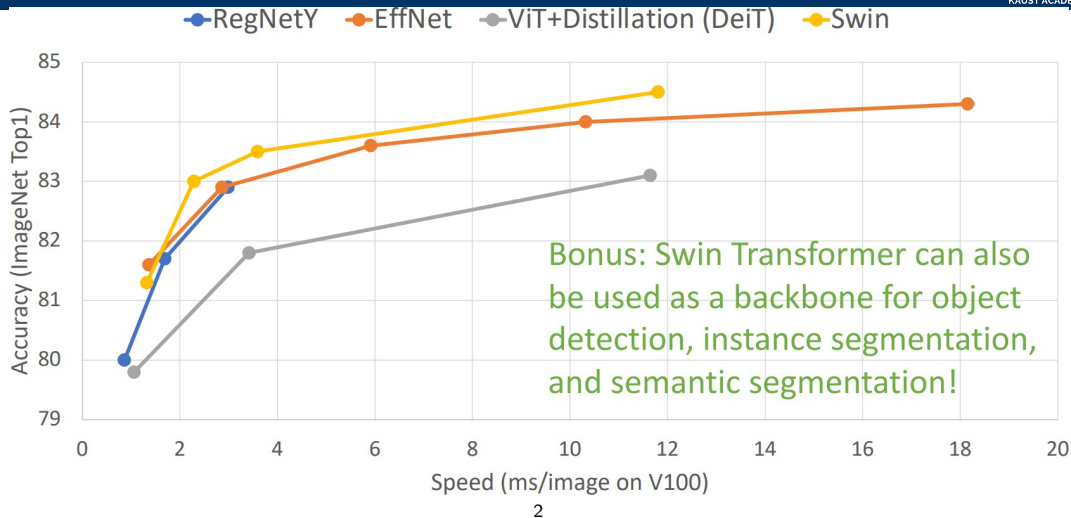
Hierarchical ViT: Swin Transformer



Source: UMich EECS 498 (2022), Lecture 18: Vision Transformers, slide 98 (after Liu et al., ICCV 2021, Fig. 3a)

- ▶ Swin Transformer introduces windowed self-attention with shifting windows, merging tokens akin to pooling.¹
- ▶ Builds a pyramid representation, enabling linear compute complexity and high performance on detection/segmentation (e.g. 58.7 COCO box AP).
- ▶ Hierarchical design allows for multi-scale feature extraction, similar to CNNs.

Hierarchical ViT: Swin Transformer (cont.)

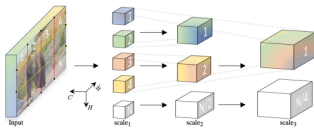


¹<https://arxiv.org/abs/2103.14030>

²Liu et al., "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows", ICCV 2021

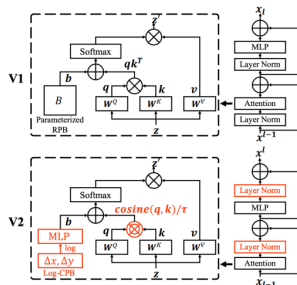
Vision Transformers: **Other Hierarchical Vision Transformers**

MViT



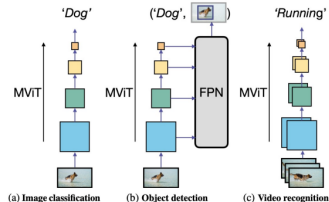
Fan et al, "Multiscale Vision Transformers", ICCV 2021

Swin-V2



Liu et al, "Swin Transformer V2: Scaling up Capacity and Resolution", CVPR 2022

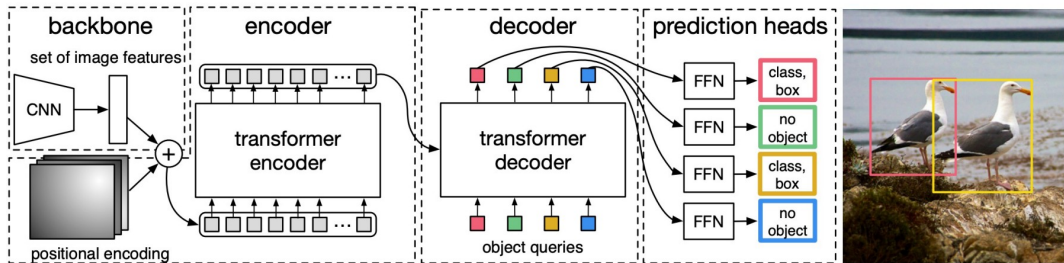
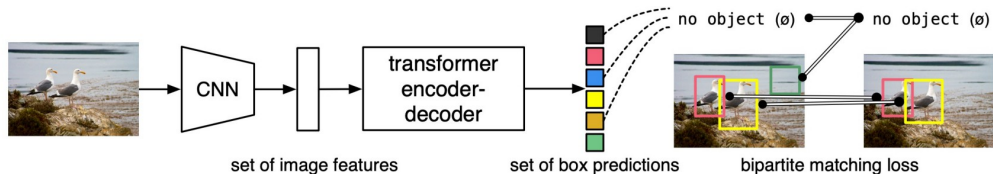
Improved MViT



Li et al, "Improved Multiscale Vision Transformers for Classification and Detection", arXiv 2021

- ▶ DETR (DEtection TRansformer, Carion et al., 2020) introduces a simple and unified object detection pipeline using Transformers.
- ▶ Transformers replace region proposal networks (RPNs) with attention-based object queries for end-to-end detection.
- ▶ DETR directly predicts a fixed set of bounding boxes and class labels, eliminating the need for anchors, hand-crafted box proposals, or non-maximum suppression (NMS).
- ▶ The approach simplifies the pipeline and enables global reasoning, though it may struggle with small objects and require higher compute.
- ▶ DETR uses bipartite matching (the Hungarian algorithm) to uniquely match predicted boxes to ground-truth boxes during training.

Object Detection with Transformers: DETR (cont.)



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³Carion et al., "End-to-End Object Detection with Transformers", ECCV 2020

<https://arxiv.org/abs/2005.12872>

- ▶ ConvNeXt is a modern CNN architecture that incorporates insights from transformer models.
- ▶ It uses a hierarchical design similar to Swin Transformer, with a focus on simplicity and efficiency.
- ▶ Key features include:
 - Depthwise separable convolutions
 - Layer normalization
 - Global average pooling
- ▶ Achieves competitive performance on image classification tasks while maintaining the strengths of CNNs.
- ▶ ConvNeXt demonstrates that CNNs can still be effective with modern design principles, even in the era of transformers.

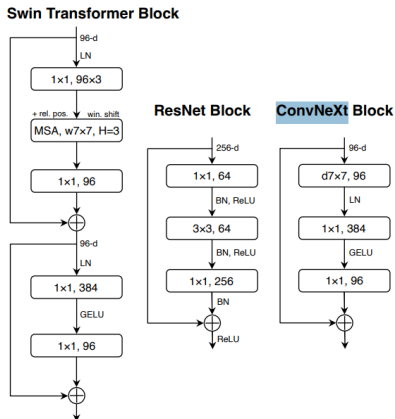


Figure 4. **Block designs** for a ResNet, a Swin Transformer, and a ConvNeXt. Swin Transformer's block is more sophisticated due to the presence of multiple specialized modules and two residual connections. For simplicity, we note the linear layers in Transformer MLP blocks also as " 1×1 convs" since they are equivalent.

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⁴Liu et al., "A ConvNet for the 2020s", CVPR 2022 <https://arxiv.org/abs/2201.03545>

Vision Transformers: **Limitations and Future Directions**

- ▶ **Data hungry:** Performance drops significantly on small datasets.
- ▶ **Computationally expensive:** Quadratic complexity of self-attention.
- ▶ **Lacks built-in locality bias:** Unlike CNNs; some variants (e.g., Swin Transformer) address this.
- ▶ **Requires large-scale pretraining:** Needs datasets like JFT-300M, ImageNet-22K for strong performance.

- ▶ **Efficient transformers:** [Linformer](#), [Performer](#), [Longformer](#).
- ▶ **Hybrid models:** Combining CNNs and Transformers (e.g., [CoaT](#), [CvT](#)).
- ▶ **Low-resource ViTs:** Data-efficient training (e.g., [DeiT](#)).
- ▶ **Vision-language models:** [CLIP](#), [Flamingo](#), [GPT-4](#).
- ▶ **Hardware acceleration:** [EdgeViTs](#), quantized ViTs for deployment.

Vision Transformers: **References**

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